PUBUDU GUNASEKARA

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MS Computer Science, Northeastern University | Seeking Software Engineering Internship, Summer 2027

EDUCATION

Northeastern University, Khoury College of Computer Sciences - San Jose, CA — Jan 2026 - May 2028 (Expected)
M.S. in Computer Science | GPA: 4.0 / 4.0
Coursework: Algorithms, Programming Design Paradigms; in progress: Machine Learning, NLP (RAG system)

Conestoga College, Postgraduate Certificate in Mobile Solutions Development | GPA: 3.74 / 4.0 — 2022 - 2023
Victoria University, Bachelor of Information Technology | GPA: 6.25 / 7.0 — 2018 - 2021

EXPERIENCE

Virtusa - Associate Engineer, QA — Jun 2021 - Mar 2022 · Colombo, Sri Lanka

- Built **Java and Selenium WebDriver test automation** for a British Telecommunications platform serving **1M+ users**, designing and executing test cases and test plans to verify system functionality across agile sprints.
- Led test planning and ran client demos and daily stand-ups for the module under the QA lead's delegation, and collaborated with developers to validate bug fixes.
- Tracked QA metrics including defect densities and open defect counts, working across CST Environment, Camunda Cockpit, RabbitMQ, Swagger UI, and MQ Console.

Gunasekara Transport - Web Developer — Dec 2025 - Jun 2026 · Remote

- Built the corporate website and web app interfaces, plus an **internal business management system** (HR, operational workflows, and reporting) that replaced manual processes, using React.js, JavaScript, PHP, WordPress, and Docker.

PROJECTS

Distributed Task Scheduler — Code · Case Study
Java 21, Spring Boot 3.5, Apache Kafka, Redis, PostgreSQL, Prometheus, Grafana, Docker, GitHub Actions

- Built a **distributed task scheduler** on a **hexagonal architecture** enforced at build time via ArchUnit, with **at-least-once delivery** over Kafka and a state-machine guard that prevents duplicate execution on redelivery.
- Implemented a **Redis distributed lock** (atomic SETNX acquire, Lua compare-and-delete release) and an exponential-backoff retry scheduler (10s, 30s, 90s) with dead-letter handling.
- Instrumented the system end-to-end with **Prometheus** metrics and **Grafana** dashboards; containerized with Docker Compose and automated build/test in GitHub Actions CI.

AI Code Review Assistant — Code · Live Demo · Case Study
React 19 / Vite, Node.js / Express, PostgreSQL, Redis, gpt-4o-mini, Docker, AWS

- Built and deployed a **full-stack app** that reviews a selected GitHub pull request via GitHub OAuth, returning severity- and category-tagged findings from gpt-4o-mini in a strict JSON schema.
- Engineered **per-user Redis rate limiting** with a fail-open policy and standard rate-limit headers, plus diff truncation to stay within the model's context budget.

Travel Day Scheduler — Code
Python, Weighted A* Search • Graduate algorithms project (team)

- Co-designed a **weighted / anytime A* itinerary planner** (team of 2) with iterative weight escalation and heap pruning to bound memory under large search spaces; implemented the initial weighted A* search and built the route-visualization frontend.

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, SQL
Backend & Distributed: Spring Boot, Node.js / Express, Apache Kafka, Redis, PostgreSQL, MongoDB, REST APIs, hexagonal architecture
Infra & DevOps: Docker, Docker Compose, GitHub Actions, CI/CD, Jenkins, Prometheus, Grafana, AWS
Frontend: React, Next.js, React Native, Vite, Redux

LEADERSHIP

- Founding Cohort Member, Graduate Student Career Advisory Board, Northeastern University (Khoury College)